Introduction to database system

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Characteristics of the Database Approach (cont'd.)

- Main characteristics of database approach
  - Self-describing nature of a database system
  - Insulationالعزل between programs and data, and data abstraction
  - Support of multiple views of the data
  - Sharing of data and multiuser transaction processing

Self-Describing Nature of a Database System

- Database system contains complete definition of structure and constraints
- **Meta-data**
  - Describes structure of the database
- Database catalog used by:
  - DBMS software
  - Database users who need information about database structure
  - A repository is a central place where information, data, files, or other resources are stored and organized so they can be accessed, managed, and retrieved when needed.

Insulation Between Programs and Data

- **Program-data independence means that a change in the structure or organization of data does not require changes to the application programs that use that data.**
  - Structure of data files is stored in DBMS catalog separately from access programs

Program-operation independence

- Program-operation independence: Program–operation independence means that an application program can use a database operation through its interface without knowing how that operation is internally implemented.
  - **Operations** specified in two parts:
    - Interface includes operation name and data types of its arguments
    - Implementation can be changed without affecting the interface

Data Abstraction

- **Data abstraction means hiding unnecessary implementation details from users and showing only the information they need.**
- **In a database system, users do not need to know how the data is physically stored. They only need to interact with the data at an appropriate level.**
  - Allows program-data independence and program-operation independence
- **Conceptual representation** of data
  - Does not include details of how data is stored or how operations are implemented
- **Data model**
  - Type of data abstraction used to provide conceptual representation

RELATIONS

Relation_name	No_of_columns
STUDENT	4
COURSE	4
SECTION	5
GRADE_REPORT	3
PREREQUISITE	2

COLUMNS

Column_name	Data_type	Belongs_to_relation
Name	Character (30)	STUDENT
Student_number	Character (4)	STUDENT
Class	Integer (1)	STUDENT
Major	Major_type	STUDENT
Course_name	Character (10)	COURSE
Course_number	XXXXNNNN	COURSE
….	….	…..
….	….	…..
….	….	…..
Prerequisite_number	XXXXNNNN	PREREQUISITE

Note: Major_type is defined as an enumerated type with all known majors.
XXXXNNNN is used to define a type with four alpha characters followed by four digits.

Figure 1.3
An example of a database catalog for the database in Figure 1.2.

Support of Multiple Views of the Data

- **View**
  - Subset of the database
  - Contains **virtual data** derived from the database files but is not explicitly stored
- Multiuser DBMS
  - Users have a variety of distinct applications
  - Must provide facilities for defining multiple views

Sharing of Data and Multiuser Transaction Processing

- Allow multiple users to access the database at the same time
- **Concurrency control software**
  - Ensure that several users trying to update the same data do so in a controlled manner
    - Result of the updates is correct
- **Online transaction processing (OLTP) application**

Sharing of Data and Multiuser Transaction Processing (cont'd.)

- **Transaction**
  - Central to many database applications
  - Executing program or process that includes one or more database
  - **Isolation** property
    - Each transaction appears to execute in isolation from other transactions
  - **Atomicity** property
    - Either all the database operations in a transaction are executed or none are

Actors on the Scene

- **Database administrators (DBA)** are responsible for:
  - Authorizing access to the database
  - Coordinating and monitoring its use
  - Acquiring software and hardware resources
- **Database designers** are responsible for:
  - Identifying the data to be stored
  - Choosing appropriate structures to represent and store this data

Actors on the Scene (cont'd.)

- **End users**
  - People whose jobs require access to the database
  - Types
    - **Casual end users**
    - **Naive** or **parametric end users**

Actors on the Scene (cont'd.)

- **System analysts**
  - Determine requirements of end users
- **Application programmers**
  - Implement these specifications as programs

Workers behind the Scene

- **DBMS system designers and implementers**
  - Design and implement the DBMS modules and interfaces as a software package
- **Tool developers**
  - Design and implement **tools**
- **Operators and maintenance personnel**
  - Responsible for running and maintenance of hardware and software environment for database system

Advantages of Using the DBMS Approach

- Controlling redundancy
  - **Data normalization**
  - **Denormalization**
    - Sometimes necessary to use **controlled redundancy** to improve the performance of queries
- Restricting unauthorized access
  - Security and authorization subsystem
  - Privileged software

Advantages of Using the DBMS Approach (cont'd.)

- Providing **persistent** storage for program objects
- Persistent means that data continues to exist after the program finishes executing.
  - Complex object in C++ can be stored permanently in an object-oriented DBMS
  - **Impedance mismatch problem**

  Impedance mismatch occurs when there is a difference between the way data is represented in a programming language and the way data is represented in a database system.
    - Object-oriented database systems typically offer data structure **compatibility**

Advantages of Using the DBMS Approach (cont'd.)

- Providing storage structures and search techniques for efficient query processing
  - **Indexes**
  - **Buffering** and **caching**
  - **Query processing and optimization**

Advantages of Using the DBMS Approach (cont'd.)

- Providing backup and recovery
  - **Backup and recovery subsystem** of the DBMS is responsible for recovery
- Providing multiple user interfaces
  - **Graphical user interfaces (GUIs)**
- Representing complex relationships among data
  - May include numerous varieties of data that are interrelated in many ways

Advantages of Using the DBMS Approach (cont'd.)

- Enforcing **integrity constraints**
  - **Referential integrity** constraint
    - Every section record must be related to a course record
  - **Key** or **uniqueness** constraint
    - Every course record must have a unique value for Course_number
  - **Business rules**
  - **Inherent rules** of the data model

Advantages of Using the DBMS Approach (cont'd.)

- Permitting inferencing and actions using rules
  - **Deductive database systems**
    - Provide capabilities for defining deduction **rules**
    - Inferencing new information from the stored database facts
  - **Trigger**
    - Rule activated by updates to the table
  - **Stored procedures**
    - More involved procedures to enforce rules

Advantages of Using the DBMS Approach (cont'd.)

- Additional implications of using the database approach
  - Reduced application development time
  - Flexibility
  - Availability of up-to-date information
  - Economies of scale

A Brief History of Database Applications

- Early database applications using hierarchical and network systems
  - Large numbers of records of similar structure
- Providing data abstraction and application flexibility with relational databases
  - Separates physical storage of data from its conceptual representation
  - Provides a mathematical foundation for data representation and querying

A Brief History of Database Applications (cont'd.)

- Object-oriented applications and the need for more complex databases
  - Used in specialized applications: engineering design, multimedia publishing, and manufacturing systems
- Interchanging data on the Web for e-commerce using XML
  - Extended markup language (XML) primary standard for interchanging data among various types of databases and Web pages

A Brief History of Database Applications (cont'd.)

- Extending database capabilities for new applications
  - Extensions to better support specialized requirements for applications
  - **Enterprise resource planning (ERP): is an integrated software system that helps an organization manage and coordinate its major business processes using a common database.**
  - **Customer relationship management (CRM)**
- Databases versus information retrieval
  - **Information retrieval (IR)**
    - Deals with books, manuscripts, and various forms of library-based articles

When Not to Use a DBMS

- More desirable to use regular files for:
  - Simple, well-defined database applications not expected to change at all
  - Stringent, real-time requirements that may not be met because of DBMS overhead
  - Embedded systems with limited storage capacity
  - No multiple-user access to data

Summary

- Database
  - Collection of related data (recorded facts)
- DBMS
  - Generalized software package for implementing and maintaining a computerized database
- Several categories of database users
- Database applications have evolved
  - Current trends: IR, Web